

1 Agent Tasks

Multi-step agent tasks with different workloads and workflows



Forward

Agent Workloads



Coding



SQL



Web Search



Bash

...

Agent Workflows



ReAct



Plan-Solve



AgentFlow

...

2 Task Scheduling



Feature Extraction

- Workload, Workflow
- Current step
- Prompt keywords



History-based Estimator

- Record running history
- Predicted remaining steps
- Estimated remaining time



Priority Scheduler

- Shorter remaining time first

4 Inference Engine

Non-SD Mode

CUDA Graph A

Autoregressive Generation

t1

Target Model (Shared)

t2

Continuous Batching



Request From Scheduler



Running Batch

Metrics

Metrics

Switch

SD Mode

CUDA Graph B

Draft + Parallel Verify

t1

Drafter

t2

t3

t4

Target Model (Shared)

t2

t3

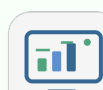
t4

3 Adaptive SD Controller



Offline Profiler

- Measure SD speedup
- Find beneficial BS range



Online Load Monitor

- Running requests BS
- Resume requests
- Waiting Queue Length



SD Switch Controller

- Enable SD when beneficial
- Disable SD under high load